

fn sample_bernoulli_exp

Michael Shoemate (@Shoeboxam)

May 5, 2026

Proves soundness of `sample_bernoulli_exp` in `mod.rs` at commit `0be3ab3e6` (outdated¹).
`fn sample_bernoulli_exp` returns a sample from the $Bernoulli(\exp(-x))$ distribution for some rational, non-negative x . This proof is an adaptation of [subsection 5.1](#) of [CKS20].

1 Hoare Triple

Precondition

Compiler-verified

- Argument x is of type `RBig`, a bignum rational

Caller-verified

- $x > 0$.

Pseudocode

```
1 def sample_bernoulli_exp(x) -> bool:
2     while x > 1:
3         if sample_bernoulli_exp1(1): #
4             x -= 1
5     else:
6         return False
7     return sample_bernoulli_exp1(x) #
```

Postcondition

Theorem 1.1. For any setting of the input parameters x such that the given preconditions hold, `sample_bernoulli_exp` either returns `Err(e)` due to a lack of system entropy, or `Ok(out)`, where `out` is distributed as $Bernoulli(\exp(-x))$.

Assume the preconditions are met.

Lemma 1.2. `sample_bernoulli_exp` only returns `Err(e)` when there is a lack of system entropy.

Proof. In all invocations of `sample_bernoulli_exp1`, the argument passed satisfies its definition preconditions, by the preconditions on x and function logic. Thus, by its definition, `sample_bernoulli_exp1` only returns an error when there is a lack of system entropy. The only source of errors in `sample_bernoulli_exp` is from the invocation of `sample_bernoulli_exp1`. Therefore `sample_bernoulli_exp` only returns `Err(e)` when there is a lack of system entropy. $\square$

Lemma 1.3. `out` is distributed as $Bernoulli(\exp(-x))$.

¹See new changes with `git diff 0be3ab3e6..7670e49 rust/src/traits/samplers/cks20/mod.rs`

Proof. For $1 \leq i \leq \lfloor x \rfloor$, let b_i denote the i^{th} outcome of `sample_bernoulli_exp1` on line 3. By the definition of `sample_bernoulli_exp1`, under the established conditions and preconditions, each B_i is distributed as $Bernoulli(\exp(-1))$. Similarly as before, C is distributed $Bernoulli(\exp(-(x - \lfloor x \rfloor)))$.

$$\begin{aligned}
P[\text{out} = \top] &= P[B_1 = B_2 = \dots = B_{\lfloor x \rfloor} = C = \top] && \text{out is only } \top \text{ if } \forall i, B_i = \top \text{ and } C = \top \\
&= \prod_{i=1}^{\lfloor x \rfloor} P[B_i = \top] P[C = \top] && \text{all } B_i \text{ and } C \text{ are independent} \\
&= \exp(-1)^{\lfloor x \rfloor} \exp(\lfloor x \rfloor - x) \\
&= \exp(-x)
\end{aligned}$$

Therefore, `out` is distributed as $Bernoulli(\exp(-x))$. □

Proof. 1.1 holds by 1.2 and 1.3. □

References

- [CKS20] Clément L. Canonne, Gautam Kamath, and Thomas Steinke. The discrete gaussian for differential privacy. *CoRR*, abs/2004.00010, 2020.